Problem Statement 4

LLM-Guided Decision Heuristics for SAT-Based Automatic Test Pattern Generation

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1. Background

Automatic Test Pattern Generation (ATPG) is a fundamental problem in digital circuit testing. Modern ATPG techniques increasingly rely on Boolean Satisfiability (SAT) solvers to generate test vectors for structural fault models such as stuck-at and transition faults.

In SAT-based ATPG, the test generation problem is encoded as a CNF formula:

- A fault-free circuit model
- A faulty circuit model
- A miter enforcing output divergence

A SAT solver is then used to determine whether a test vector exists.

While SAT-based ATPG has shown strong performance improvements over classical algorithms (e.g., PODEM, FAN), the efficiency of SAT solving heavily depends on internal heuristics such as:

- Decision variable ordering
- Clause learning
- Branching strategies

These heuristics are purely syntactic and unaware of circuit semantics.

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2. Problem Gap

Current SAT-based ATPG approaches:

- Treat each fault independently
- Use generic SAT heuristics (e.g., VSIDS)
- Do not leverage structural or functional circuit knowledge beyond CNF encoding
- Do not learn semantic patterns across faults

Large Language Models (LLMs), however, are capable of reasoning over:

- Circuit structure

- Gate-level dependencies
- Sensitization paths
- Functional behavior descriptions

There is currently little to no systematic study on:

Whether LLMs can serve as semantic heuristic oracles to guide SAT-based ATPG search and improve solving efficiency.

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3. Problem Statement

The objective of this project is to design, implement, and evaluate an LLM-guided SAT-based ATPG framework where the LLM assists the SAT solver by providing structural decision heuristics for fault activation and propagation.

Specifically, the project must address the following research questions:

1. Can an LLM predict useful partial assignments for fault activation and propagation?
2. Can LLM-guided variable selection reduce SAT solver:
    - Decision count?
    - Conflict count?
    - Solving time?
3. Does LLM guidance improve detection of hard-to-detect faults?
4. Can knowledge transfer across previously solved faults improve performance on new faults?

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4. Proposed Approach

Students must implement the following architecture:

Step 1: SAT-Based ATPG Core

- Convert RTL to gate-level netlist (using Yosys or equivalent)
- Encode stuck-at faults in CNF
- Construct good/faulty circuit miter
- Solve using a SAT solver (e.g., MiniSAT / PySAT / Z3)

Step 2: LLM Guidance Layer

For each fault instance, the LLM may provide one or more of the following:

- Predicted fault activation conditions
- Suggested sensitization paths
- Partial input assignments
- Variable ordering hints

These hints must be translated into:

- Assumption literals
- Branching priority modifications
- CNF constraints (soft guidance)

Step 3: Closed-Loop Evaluation

If SAT fails or conflicts heavily:

- Extract UNSAT core or conflict information
- Provide structured feedback to LLM
- Refine guidance iteratively

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5. Expected Novelty

The novelty of this project lies in:

1. Introducing semantic reasoning from LLMs into SAT-based ATPG decision processes.
2. Formally evaluating LLM impact on solver complexity metrics.
3. Studying hybrid semantic-structural heuristics in ATPG.
4. Investigating cross-fault knowledge reuse via LLM interaction.

This is distinct from classical SAT-ATPG work, which relies solely on algorithmic heuristics internal to the solver.

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6. Experimental Evaluation

Students must evaluate their framework on benchmark circuits such as:

- ISCAS85 / ISCAS89
- Arithmetic modules (ALU, multipliers)

Metrics to report:

- Fault coverage
- SAT solving time
- Number of decisions
- Conflict count
- Pattern count
- Comparison with baseline SAT-ATPG

Statistical comparison must be included.

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7. Deliverables

1. SAT-based ATPG implementation
2. LLM integration module
3. Experimental evaluation report
4. Complexity analysis
5. Discussion on scalability and limitations

Bonus (optional):

- Extension to transition faults
- LLM-guided test point insertion
- Learning across multiple circuits

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8. Expected Learning Outcomes

Students will gain:

- Practical understanding of SAT-based ATPG
- Experience in CNF encoding of hardware circuits
- Insight into solver heuristics
- Critical evaluation of AI-assisted formal reasoning
- Ability to distinguish hype from measurable algorithmic improvement

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If you would like, I can also:

- Convert this into a polished PDF-ready project handout
- Add grading rubric (research depth vs implementation vs evaluation)
- Or provide a weaker B.Tech version and a stronger M.Tech/PhD version

This has the potential to be a genuinely strong formal methods course project.